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# InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference

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***Abstract*—The widespread of Large Language Models (LLMs) marks a significant milestone in generative AI. Nevertheless, the increasing context length and batch size in offline LLM inference escalate the memory requirement of the key-value (KV) cache, which imposes a huge burden on the GPU VRAM, especially for resource-constrained scenarios (e.g., edge computing). Several cost-effective solutions leverage host memory or SSDs to reduce storage costs for offline inference scenarios and improve the throughput. Nevertheless, they suffer from significant performance penalties imposed by intensive KV cache accesses due to limited PCIe bandwidth. To address these issues, we propose *InstAttention*, a novel LLM inference system that offloads the most performance-critical computation (i.e., attention in decoding phase) and data (i.e., KV cache) parts to Computational Storage Drives (CSDs), which minimize the enormous KV transfer overheads. InstAttention designs a dedicated flash-aware in-storage attention engine with KV cache management mechanisms to exploit the high internal bandwidths of CSDs instead of being limited by the PCIe bandwidth. The optimized P2P transmission between GPU and CSDs further reduces data migration overheads. Experimental results demonstrate that for a 13B model using an NVIDIA A6000 GPU, InstAttention improves throughput for long-sequence inference by up to 11.1×, compared to existing SSD-based solutions such as FlexGen.**

## I. Introduction

Large language models (LLMs) and their underlying transformer architecture have revolutionized AI and have become the bedrock of many emerging applications, widely used in domains such as chatbot [2], summarization [64], and code generation [47]. Most of these LLMs are built based on the transformer architecture [62] with an enormous number of parameters and perform inference in an autoregressive manner consisting of two phases: *prefilling* phase and *decoding* phase.

Previous research [3], [20], [67], [80] indicate that the prefilling phase is compute-bound, while the decoding phase turns memory bound due to its key technique *KV cache*. It stores intermediate key and value tensors from previous tokens, significantly reducing computational complexity by allowing the model to reference past information efficiently without time-consuming recomputations [16], [32]. Considering the computing and bandwidth requirements, leveraging the extensive computing power and large bandwidth of GPU to accelerate LLM inference is the mainstream choice. As illustrated in Figure 1(a), the `GPU-only` architecture stores

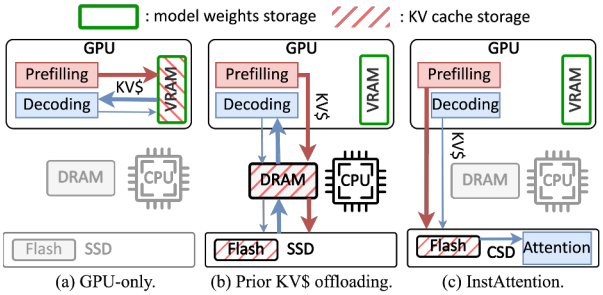


Fig. 1: Comparison of different LLM inference architectures.

all weights and KV caches in the VRAM and leverages the GPU to accelerate both prefilling and decoding phases.

Current LLM inference services can be categorized into online and offline scenarios. Online inference prioritizes low latency and typically accepts shorter sequences from users [15]; whereas offline reasoning usually deals with longer sentences, accepting longer delays in exchange for higher throughput [57]. As LLMs continue to evolve and push the boundaries toward longer context reasoning [26], [37] and larger batches [28], [57], the memory footprint of their associated KV cache in offline-inference escalates drastically [33], introducing substantial challenges in storing them efficiently. The situation gets more severe in resource-constrained scenarios such as edge computing or personal devices [5], [35], [74]. To be specific, the financial burden of deploying additional GPUs to accommodate the expansive KV cache can become exorbitantly high, potentially exceeding even the costs associated with storing the model weights. For instance, a mid-sized LLM with 13 billion parameters, operating at a batch size of 32 and 4K tokens, necessitates approximately 100GB of KV cache. This volume is 4.2× the size of the model itself.

To mitigate the storage costs associated with the KV cache, several approaches (e.g., DeepSpeed-MII [21] and FlexGen [57]) have adopted more economical solutions, which offload the KV cache to host memory or cheaper SSDs for throughput-oriented offline inference, as shown in Figure 1(b). Before the GPU begins the decoding phase of inference, the KV cache is first loaded from SSDs to the memory and then to the GPU via PCIe buses. However, this offloading strategy introduces severe performance penalties. In particular, the PCIe bandwidth

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between host memory and GPUs is substantially lower than the bandwidth within GPU VRAM [33], while the bandwidth of SSDs is even lower. Additionally, the lack of direct datapath between the SSD and GPU and the complicated host-oriented storage software stack further exaggerate the performance penalty of SSD-offloading solutions. Unlike the compute-bound prefilling phase, the memory-bound decoding phase critically depends on KV cache I/O, as it requires frequent transfers of large KV cache volumes between the storage media and GPUs. This dependence makes data movement over a narrow PCIe bus a new performance bottleneck.

To address the storage cost and bandwidth issues associated with KV cache, Computational Storage Drives (CSDs) [34], [42], [72] become a promising and cost-effective solution. Built on modern high-capacity SSDs, CSDs integrate computational resources such as FPGA accelerators internally. They present two advantages: 1) The storage cost of CSDs is comparable to that of SSDs [8], [30]. Unlike the expensive GPU and DRAM, the affordable storage capacity of SSDs can satisfy substantial capacity requirements of KV cache for long-context and large-batch scenarios. 2) Modern SSDs aggregate the throughput of all flash chips to deliver high internal bandwidth (tens of GB/s) [50], [66], [76], which is significantly higher than the external PCIe bandwidth (3$\sim$7 GB/s) [17], [46]. Offloading inference to the computing engines in CSDs allows operands to leverage the high internal bandwidth directly. This bypasses the bandwidth-limited external PCIe bus, thereby meeting the KV cache bandwidth requirements.

Nevertheless, due to power consumption and cost constraints [18], the computational power of CSDs is 2$\sim$3 orders of magnitude weaker than GPUs, making it ineffective to accelerate the entire inference tasks (cf. Section III-B). Instead, CSDs must collaborate with GPUs to accelerate LLM inference as a novel heterogeneous system, which, while seemingly straightforward, presents significant challenges:

• *Coarse task partitioning between the GPU and CSD.* Existing heterogeneous LLM inference solutions typically disaggregate the prefilling and decoding phases [52], [80]. However, considering the much lower computing power of the CSD compared to the GPU, the entire decoding task exceeds the computing capability of the CSD, which becomes a new performance bottleneck.

• *Significant bandwidth gap between CSD and GPU.* Both the external and internal bandwidth of CSD are still much lower than the GPU. For memory-bound decoding phase inference, reducing the data migration overheads remains necessary.

• *Discrepancy between flash and memory access patterns.* NAND flash accesses necessitate page granularity, high access latency, and complex multi-layer address translation mechanisms including the host file system and the flash Translation Layer (FTL) [23]. Therefore, existing KV cache management mechanisms designed for memory (e.g., vLLM [32]) cannot be directly applied within the CSD.

Tackling the aforementioned challenges, we propose *InstAttention* [1], a novel LLM inference system based on in-storage computing and flash-based KV cache offloading, which effectively addresses both storage cost and bandwidth limitations in traditional offline-inference schemes incurred by enormous KV cache volume, as illustrated in Figure 1(c). Specifically, to alleviate the computing burden of CSDs, InstAttention only offloads the most performance-critical *decoding-phase attention* computations during long-context inference to CSDs, while leveraging the GPU to execute the remaining inference tasks. To mitigate the computation power and bandwidth gap between CSD and GPU, InstAttention designs dedicated flash-aware in-storage computation engines with algorithm-hardware co-design for attention operators, which effectively lower the computation intensity and KV cache demands. InstAttention further proposes a KV cache-oriented FTL design to enable efficient KV cache access on the flash chips. The GPU and CSDs are directly connected via PCIe peer-to-peer DMA [48], bypassing the host to avoid extra copies. Meanwhile, the works as the control plane, which only manages user requests, task scheduling, and data movement coordination. To the best of our knowledge, InstAttention is the *first work* to exploit *CSDs* to address the performance penalty incurred from KV cache offloading. Experimental results show that for a 13B model with an NVIDIA A6000 GPU [49], the throughput for long-sequence inference is improved by up to 11.1$\times$, compared to FlexGen.

The main **contributions** of this work are as follows:

• *Pioneering CSD-based or GPU-CSD heterogeneous LLM inference system for long contexts:* Our detailed analysis reveals that the decoding-phase attention is the most critical performance bottleneck due to the restricted PCIe bandwidth to access large KV caches. It exhibits extremely low arithmetic intensity, rendering GPU acceleration ineffective. To address this, InstAttention offloads both KV cache and memory-intensive decoding-phase attention to the CSD, which exploits the high aggregated bandwidth of flash chips. Consequently, the data migration overheads are effectively mitigated by up to 94.0%, while the prefilling phase overheads are further alleviated by the optimized peer-to-peer DMA mechanism.

• *Hardware-algorithm co-designed in-storage attention engine:* To effectively bridge the bandwidth and computation power gap between GPU and CSD, we propose the bandwidth-efficient *SparF* algorithm, which not only reduces the computing intensity but also minimizes the required KV cache volume during the decoding-phase attention while maintaining accuracy. Considering the page granularity of flash accesses, InstAttention incorporates a *dual-step loading* strategy to manage the sparsity in the sequence: initially at the page level and subsequently at the token level. We further implement the in-storage SparF attention engine in hardware kernels, with fine-grained parallelism design to conceal the long access latency of flash chips, thereby improving the inference efficiency.

• *KV cache-oriented FTL design for efficient retrieval:* With

[1]InstAttention is open-sourced and can be accessed at https://github.com/ChaseLab-PKU/InstAttention.

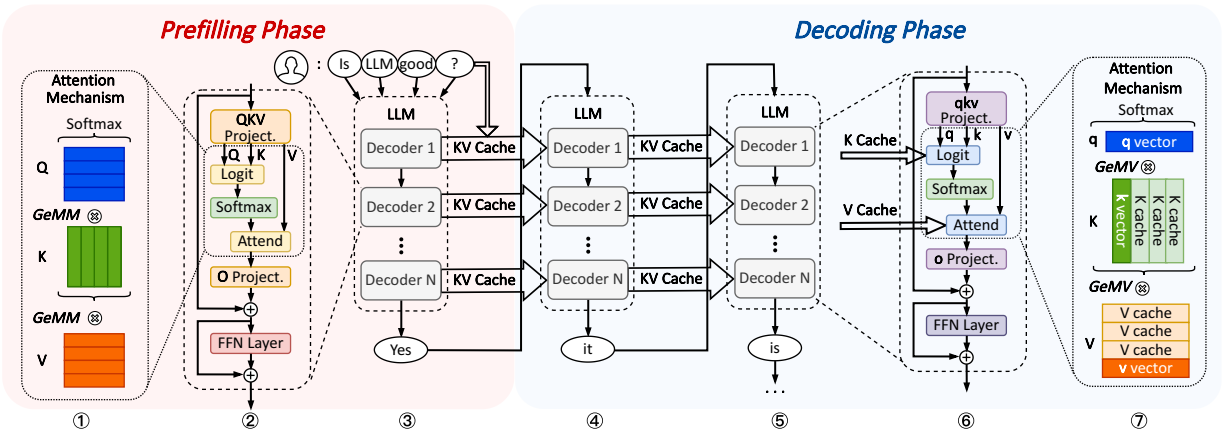



Fig. 2: General architecture of LLM and the inference flow.

the SparF algorithm identifying sparsity patterns in both tokens and channels, the resultant random access to KV caches in the flash chips presents a significant challenge. InstAttention confronts this issue by introducing *dual address mapping* mechanisms tailored for token-indexed and channel-indexed KV caches, respectively. We further meticulously organize KV cache tensors into groups that align with flash page sizes and distribute them across multiple flash blocks and chips in a stridden fashion for each attention head, thus exploiting the inherent high parallelism.

## II. Background

### A. LLM Inference Basics

**LLM Architecture.** Mainstream Large Language Models (LLMs) predominantly utilize a decoder-only transformer architecture [61], [63], [71], [78]. It primarily comprises multiple decoder blocks, each consisting of a self-attention module and a Feed-Forward Network (FFN) module. As illustrated in blocks ②,③ in Figure 2, for a given sequence of inputs $X = [x_1, ..., x_s]$, each decoder block applies linear transformations to $X$ with the parameter matrices, mapping $X$ into three embedding matrices: Q, K, and V, through GeMM computations. Subsequently, the attention mechanism [62] is performed to capture the semantic context of the sentence: $Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$. To enhance the ability of the vanilla attention mechanism to capture various aspects of the context, the Multi-Head Attention (MHA) [62] further divides the QKV matrices into multiple smaller matrices. This approach allows the model to focus on different parts of the input sequence simultaneously. The resultant attention output is then subjected to a linear transformation via the O matrix and processed by the FFN layer. This output is then fed into the next decoder block as input. After all the decoder blocks, the final predicted token is generated.

**Auto-regressive Inference.** LLM inference leverages an auto-regressive approach [33], consisting of the prefilling phase and the decoding phase (cf. blocks ③∼⑤ in Figure 2). During prefilling, the LLM processes all the tokens of the input prompts in parallel to generate the first predicted output token $x_{s+1}$. This token is then appended to the existing input prompt sequence to generate the new input sequence $[x_1, ..., x_s, x_{s+1}]$. When decoding, the LLM predicts one new output token at a time based on this sequence, and gets the predicted token $x_{s+2}$. This process repeats iteratively until an End-of-Sequence token is generated or the model reaches its context limit.

### B. KV Cache

**Recomputation reduction.** During the decoding phase of LLM inference, the input for each inference step consists of the entire sequence generated so far. Consequently, the attention operation requires repeated calculations of the QKV matrices for all the previous tokens, resulting in a computational complexity of $O(s^2)$ per iteration [62].

An effective method to alleviate the computational bottleneck in LLM decoding is the KV cache [32] (cf. blocks ⑥ and ⑦ in Figure 2). By caching the KV tensors for the generated tokens in the GPU VRAM, redundant calculations can be avoided. Thus, when computing the new attention output, only the KV vectors for the new token need to be calculated. This optimization reduces the attention calculation in each decoding step from GeMM to GeMV, thereby lowering the computational complexity of the attention layer from $O(s^2)$ to $O(s)$. However, as the context length of LLMs increases, storing the KV cache consumes substantial memory space and imposes high I/O demands during the decoding phase [33].

**Sparse Attention.** To further reduce the memory access demands of attention, sparse attention has become a commonly adopted method [9], [10]. This approach is based on the observation that within a text sequence, the importance of different tokens varies; in a fully connected attention mechanism, some weak connections contribute minimally to the final attention output and can be disregarded. By reducing the number of KV vectors for tokens to be calculated and stored, sparse attention opens up possibilities for decreasing the computational and storage overhead of the KV cache.

Prior works have proposed various sparse attention algorithms [38], [39], [54], [68], [79], among which SparQ Attention [54] is optimized for bandwidth-efficient inference

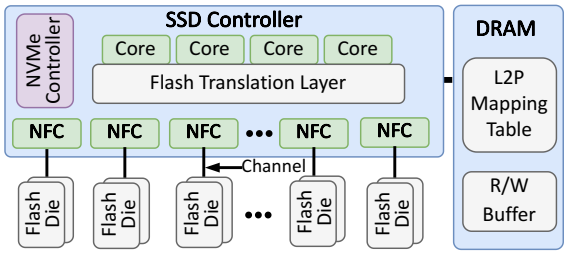



Fig. 3: A typical SSD architecture.

scenarios. Unlike other algorithms that compute the complete attention score, it approximates the score based on the $r$ largest entries in the query (Q) vector. It then identifies the top-$k$ most important tokens based on the approximated attention score with full hidden embeddings to calculate the attention output. To compensate for the omitted value (V) tensors, the V tensors are weighted and averaged, merging them into the final attention output. On multiple datasets, SparQ Attention reduces the bandwidth requirement for KV cache transmission during the decoding phase by up to 7/8 while maintaining accuracy (cf. Section VI-B). However, the SparQ attention algorithm only reduces the bandwidth demand for KV cache access but requires 1.5× larger KV cache memory footprints. This is because it needs to index the key (K) cache by both token dimension and channel dimensions, limiting its applicability in memory-constrained scenarios.

### C. SSD and In-Storage Computation

**SSD Basics.** Figure 3 illustrates the internal organization of a modern NAND-flash-based solid-state drive (SSD) [43], which comprises three main components: NAND flash dies, an SSD controller, and a DRAM module. One or more dies share command/data buses, known as *channels*, to connect to the SSD controller. Each die is subdivided into 2∼4 planes, and each plane contains thousands of blocks. A block is further divided into hundreds of pages, typically ranging from 4KB to 16KB in size [12]. Pages are the smallest read/write units of flash chips. Before data can be written to flash pages, the flash memory needs to be erased at the block level [4].

The SSD controller generally consists of three parts: a general-purpose processor running the flash translation layer (FTL), an NVMe controller, and NAND flash controllers (NFCs). The FTL is responsible for managing the logical-to-physical address mapping of the data stored in the flash dies and scheduling tasks on the NAND flash. The NVMe controller facilitates communication with the host via the NVMe protocol [14], while the NFCs manage communication with the flash backend. Each NFC operates on a flash channel for independent data transfers. Modern SSDs typically feature 8∼16 flash channels, with each channel capable of transferring data at rates of 1∼2GB/s [76]. Consequently, the aggregated bandwidth of flash channels can reach tens of GB/s, significantly exceeding the external PCIe bandwidth of SSDs (3∼6GB/s) [55]. The DRAM within the SSD functions as a temporary buffer for data being read from or written to the flash dies. It also maintains the logical-to-physical (L2P) mapping table and other metadata for the FTL.

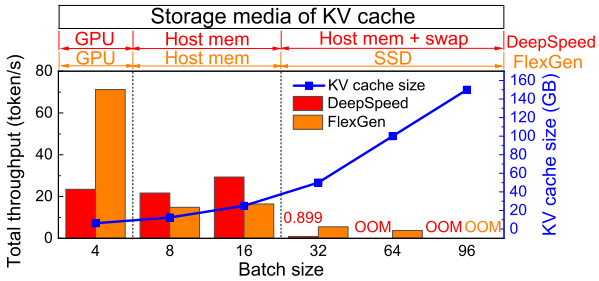



Fig. 4: Throughput of DeepSpeed and FlexGen.

**Computational Storage Drive.** To leverage the high internal bandwidth of modern SSDs, the computational storage drive (CSD) employs in-storage computation techniques by integrating computing engines, such as ARM cores, NPUs, or FPGA chips, within the SSD [30], [36], [42]. This integration endows the SSD with computing capabilities, enabling it to perform data processing tasks directly within the storage device. It is worth noting that, to fully utilize the high flash channel bandwidth, it would be better to place the computing engine near the flash dies or NFCs rather than being connected to the SSD through a PCIe switch (i.e., Samsung SmartSSD [58]). This in-storage computing architecture is employed in InstAttention to harness the substantial bandwidth necessary for LLM inference with a large KV cache.

## III. Challenges and Opportunities

### A. Limitations of Conventional KV Cache Offloading

**KV Cache Analysis.** Nowadays, the context length of the LLM inference serving system is continuously increasing [26], [37].Furthermore, as LLMs become prevalent, both the user base and usage frequency significantly increase, leading to more concurrent inference requests for LLM servers. Consequently, for resource-constrained scenarios [40], [70], [77] such as offering inference services to a small group of people at the edge or a large number of users at medium-sized LLM Agent servers, one common practice to enable cost-effective LLM inference systems is to batch many requests in a single iteration, which can effectively enhance the GPU utilization rates. However, all the above lead to the KV cache capacity bloat. Assuming that $b, s, p$ denote the batch size, sequence length, and model size, respectively, the KV cache size stored in the FP16 format is $4bsp$, while the model size in FP16 format is only $2p$. For a 2K-length sequence with $b = 128$, the OPT-13B model occupies about 24GB for its model weights and generates 200GB KV caches. For larger models like OPT-175B, the model weights occupy 325GB, while the KV cache reaches up to 2.63TB. Given that the precious GPU memory will be primarily allocated for storing the model weights and activations, the KV cache tends to be offloaded to host memory or SSDs for cost-effectiveness, depending on the sizes.

**Performance Degradation With Offloading.** As the PCIe bandwidth between host memory or SSDs and the GPU is

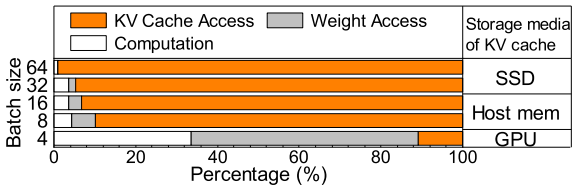



Fig. 5: Latency breakdown of FlexGen decoding.

significantly lower than that of GPU memory, prior offloading schemes lead to a noticeable decline in inference performance. To demonstrate this, we evaluate the inference throughput of two latest KV cache offloading systems, DeepSpeed [21] and FlexGen [57], in a long-context scenario of the OPT-13B model. We evaluate them with different batch sizes on an NVIDIA A6000 GPU, which possesses 48GB GPU memory. Both the input and output sequence lengths are set to 1024 tokens. As depicted in Figure 4, both Deepspeed and FlexGen exhibit performance drop as batch sizes increase: Deepspeed at batch sizes 8 and 32, and FlexGen at batch sizes 8 and 64. These drops occur because the KV cache size exceeds the available GPU memory, necessitating offloading first to host memory and subsequently to SSD. Note that Deepspeed does not support SSD offloading; consequently, at a batch size of 32, kernel swapping from host memory to SSD occurs, leading to a 97.01% performance decline. While increasing the batch size within the same memory tier enhances throughput, offloading the KV cache to secondary storage significantly degrades the performance.

To further elucidate the source of the performance penalty, we analyze the decoding-phase latency of FlexGen across different batch sizes, as illustrated in Figure 5. For smaller batch sizes (4, 8), where all the KV caches fit within the GPU, the primary bottleneck is `Weight Access`. However, as the batch size increases and the KV caches are offloaded to memory or SSD, the overhead from `KV Cache Access` escalates to as high as 98.94%. This substantial increase underscores the need for new solutions to address the significant performance challenges introduced by KV cache offloading.

### B. Offloading Opportunities with CSD

We discovered that compared to memory and NVMe SSDs, offloading KV caches to the flash chips within the CSD can directly leverage the higher flash channel bandwidth to meet the demands of the LLM decoding phase using the internal computational units. However, the simple prefilling-decoding separation architecture proposed in prior works [52], [80], which typically targets GPU-CPU separation or distribution across different GPUs, is not suitable for CSD offloading. This is primarily due to the significant differences in the characteristics of various operators and the much lower performance of CSD compared to GPUs. Consequently, it is challenging for CSDs to handle the entire inference task independently.

To minimize the computational load on the CSD and fully utilize its high internal bandwidth, a practical approach involves restructuring the scheme of task disaggregation. This

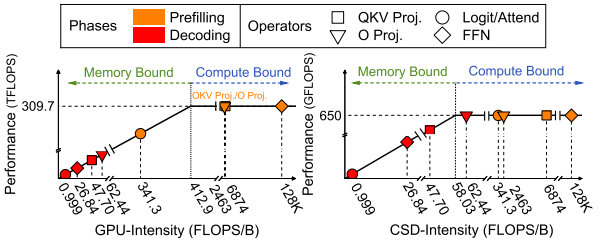



Fig. 6: Roofline models of different hardware.

can be achieved by offloading only memory-bound operators with low computing intensity and substantial KV cache I/O to the CSD, while retaining other operators on the GPU.

To this end, we thoroughly analyzed the main operators in LLM inference, examining their patterns on both CSD and GPU. Figure 6 illustrates the roofline models [75] of an NVIDIA A6000 GPU [49] and a Zynq7045 FPGA-based CSD [69]. The hardware configurations are detailed in Section V. For the prefilling phase, `QKV Proj.`, `O Proj.`, and `FFN` are extremely computing-intensive and should be placed on GPU. Although the attention operands (i.e., `Logit` and `Attend`) are memory-bound on the GPU, the limited computing power on CSD will severely constrain their performance. Therefore, the prefilling-phase attention should also remain on the GPU.

In contrast, the decoding-phase operators exhibit significantly different characteristics. Although `QKV Proj.`, `O Proj.`, and `FFN` operands are memory-bound on the GPU and seem suitable for CSD-offloading, their operational intensities are near the maximum computing capability of CSD. This places a substantial burden on CSD's computing engine. Moreover, these operands rely solely on weight matrices for flat GeMM computations [22], independent of the KV cache on the flash chips. Conversely, the attention operands (`Logit` and `Attend`), which involve extremely low-intensity GeMV computations with 1:1 computation-memory-access ratio, require direct access to KV caches (see block ⑦ in Figure 2). Considering the maximum 650GFLOPS computation capacity of CSD, the theoretical throughput requirements would be 650GB/s, which is hard to reach for general SSDs over PCIe lanes (3∼10GB/s). This motivates us to offload the decoding-phase attention operands to the CSD while retaining other processes on the GPU. This approach aims to significantly reduce KV cache transmission overheads and minimize the computational burden on the CSD.

## IV. Design

### A. Overview of InstAttention

Based on the insights presented in Section III, we propose *InstAttention*, the first in-storage attention offloading system with general GPUs, tailored for offline LLM inference with long-context and large-batch.

The key idea of InstAttention lies in reducing both KV cache movement overheads and computational burden on the CSD, along with the corresponding flash-aware KV cache retrieval mechanism and co-design of attention operands. As

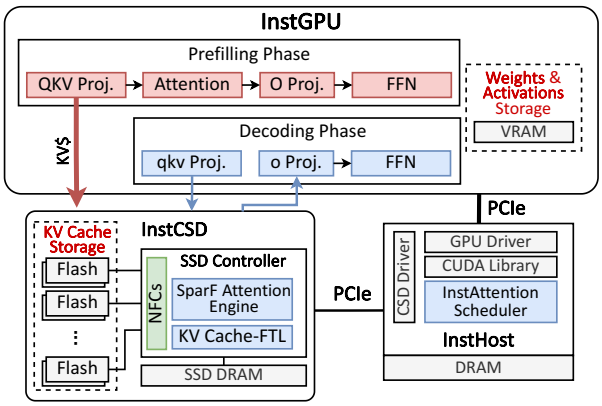



Fig. 7: Overview of InstAttention architecture.

illustrated in Figure 7, InstAttention is primarily comprised of three hardware components: 1) *InstCSD*, which executes decoding-phase attention computation and stores the large KV cache volumes; 2) *InstGPU*, which performs other inference computations along with generating KV cache during the pre-filling phase; and 3) *InstHost*, which runs the software stack, scheduling inference tasks and orchestrating data transmission between the GPU and InstCSDs.

Given that for KV cache in the CSD, the storage requirement is significantly less than the demanding bandwidth, we propose the *SparF Attention* mechanism, an enhanced version to the traditional SparQ algorithm [54] (cf. Section II-B), specifically tailored for flash storage to trade storage capacity for reduced computation and data transmission on the CSD. Considering the page granularity of flash access, SparF Attention organizes tokens at a group level, which corresponds to the page size of flash chips to avoid wasting the flash channel bandwidth. KV cache tensors are identified and fetched via a dual-step mechanism, initially at the coarse-grained group level and then at the fine-grained entry level.

Based on the SparF Attention, we further design the hardware-based accelerator on InstCSD via FPGA, which computes the attention outputs at fine-grained parallelism, to effectively identify sparsity patterns in runtime. As SparF requires to index KV cache in both channel and token, we propose two address-mapping mechanisms in the FTL of InstCSD for efficient KV cache retrieval. Through the InstCSD, only $qkv$ vectors and attention output are transmitted between the GPU and CSD during the decoding phase. Furthermore, the KV caches generated by the GPU are transmitted to CSD through P2PDMA, bypassing the host memory and the burden from the filesystem. The KV cache transmission is executed in a layerwise way, overlapped with the inference computation to hide the transmission latency. Note that as LLMs and the corresponding optimization techniques are experiencing rapid evolution, we aim to explore an architectural solution for LLM acceleration through FPGA-based CSD rather than focusing on a specific algorithm. Considering LLM pruning techniques are rapidly evolving, InstAttention adopts FPGA-based acceleration units to serve as a flexible solution for various algorithms.

**Algorithm 1** SparF Attention: flash-aware Sparse q-Attention

**Input:** $\boldsymbol{q}, \bar{\boldsymbol{v}} \in \mathbb{R}^{d_h}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{S \times d_h}, \boldsymbol{r}, \boldsymbol{k} \in \mathbb{N}$
**Output:** $\boldsymbol{out} \in \mathbb{R}^{d_h}$

$\boldsymbol{i} \leftarrow [1 \textbf{ if } i \in \text{argtopk}(|\boldsymbol{q}|, r) \textbf{ else } 0]_{i=1}^{d_h}$
**for** $i_1 = 1$ **to** $num_{group}^{ch}$ **do**
  **if** all entries in $group_{i_1}$ are zero **then**
    load $\boldsymbol{K}_{[:,i_1]}^{\top}$ {*SparF Filter-1*}
    $\boldsymbol{K}_{[:,i]}^{\top} \leftarrow \boldsymbol{K}_{[:,\boldsymbol{i}_2]}^{\top}$ **for** $\boldsymbol{i}_2 \neq 0$ **in** $group_{i_1}$ {*SparF Filter-2*}
  **end if**
**end for**
$\hat{\boldsymbol{s}} \leftarrow \text{softmax}\left(\boldsymbol{q}_{[\boldsymbol{i}]} \cdot \boldsymbol{K}_{[:,\boldsymbol{i}]}^{\top} / \sqrt{d_h \frac{||\boldsymbol{q}_{[\boldsymbol{i}]}||_1}{||\boldsymbol{q}||_1}}\right)$
$\boldsymbol{m} \leftarrow [1 \textbf{ if } i > S \textbf{ else } 0]_{i=1}^{S}$
$\boldsymbol{j} \leftarrow [1 \textbf{ if } j \in \text{argtopk}(\hat{\boldsymbol{s}} + \boldsymbol{m}, k) \textbf{ else } 0]_{j=1}^{S}$
$\alpha \leftarrow \text{sum}(\hat{\boldsymbol{s}}_{[\boldsymbol{j}]})$
**for** $j_1 = 1$ **to** $num_{group}^{tk}$ **do**
  **if** all entries in $group_{j_1}$ are zero **then**
    load $\boldsymbol{K}_{[\boldsymbol{j}_1,:]}^{\top}, \boldsymbol{V}_{[\boldsymbol{j}_1,:]}$ {*SparF Filter-3*}
    $\boldsymbol{K}_{[\boldsymbol{j},:]}^{\top}, \boldsymbol{V}_{[\boldsymbol{j},:]} \leftarrow \boldsymbol{K}_{[\boldsymbol{j}_2,:]}^{\top}, \boldsymbol{V}_{[\boldsymbol{j}_2,:]}$ **for** $\boldsymbol{j}_2 \neq 0$ **in** $group_{j_1}$ {*SparF Filter-4*}
  **end if**
**end for**
$\boldsymbol{s} \leftarrow \text{softmax}\left(\boldsymbol{q} \cdot \boldsymbol{K}_{[\boldsymbol{j},:]}^{\top} / \sqrt{d_h}\right)$
$\boldsymbol{out} \leftarrow \alpha \boldsymbol{s} \cdot \boldsymbol{V}_{[\boldsymbol{j},:]} + (1 - \alpha)\bar{\boldsymbol{v}}$

### B. Compute Attention Outputs

**Flash-Aware Sparse Attention.** Based on Section III-B, we observed that the decoding-phase attention remains severely memory-bound on CSD due to its predominate reliance on the KV cache in the flash chips. Our solution leverages the inherent sparsity in attention, which has been thoroughly exploited in prior works (cf. Section II-B). However, prior approaches do not consider the specific flash characteristics, rendering them unsuitable for CSD adoption. Specifically, unlike memory, NAND-flash-based SSD has a much larger capacity with lower bandwidth, making it feasible to trade storage capacity for bandwidth. Furthermore, current sparsity algorithms generate extensive random accesses to the KV cache due to the varying semantic relatedness among tokens in different contexts. It leads to random accesses in flash chips, resulting in significant write amplification [24] and bandwidth wastage due to the page granularity of flash chip accesses.

To this end, we propose *SparF Attention*, a flash-aware sparse q-attention algorithm that builds on the vanilla SparQ attention [54], as delineated in Algorithm 1 assuming an LLM with a hidden dimension of $d_h$, sequence length of $S$, and batchsize=1. The enhancements specific to SparF are highlighted in Algorithm 1 with {*SparF Filters*}. SparF identifies the sparsity pattern between the current token (i.e., the $q$ vector) and existing sequence (i.e., the $K$ cache matrix) by selecting the top-$r$ entries of the $q$ vector (*step* 1 in

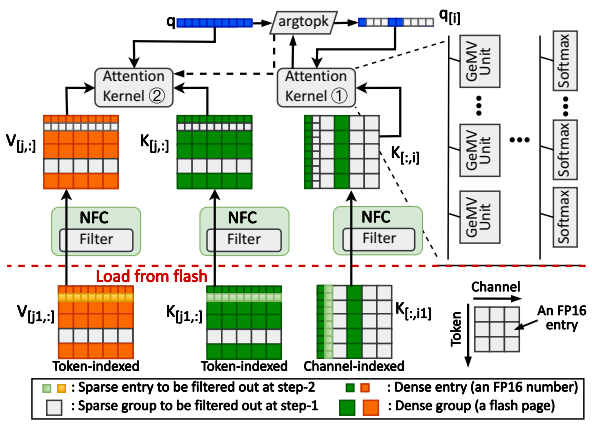



Fig. 8: Workflow and architecture of SparF engine on InstCSD.

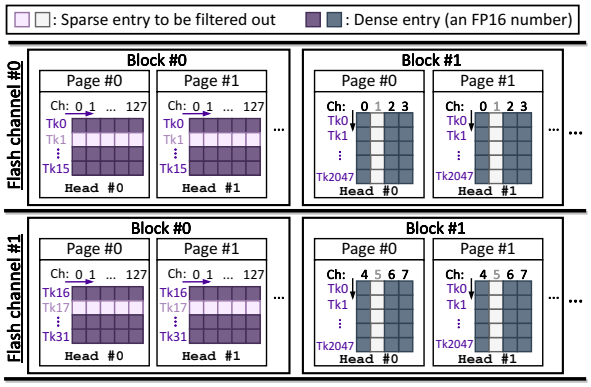



Fig. 9: Schematic of different KV cache mapping schemes.

Algorithm 1) to approximate the full attention score $\hat{s}$. This involves loading the $K$ caches from flash chips indexed by the hidden embedding channel (*steps* $2-7$), matching the identified sparsity in $q$. Subsequently, based on the approximated attention score $\hat{s}$, SparF selects the top-$k$ largest tokens from $\hat{s}$ to approximate the final output (*step* 10). It then loads the corresponding full $K, V$ cache tensors for these tokens from the flash chips (*steps* $12-17$).To fit with the flash page size, the KV cache loading process is structured into two phases, as in steps $4-5, 14-15$ in Algorithm 1. The detailed data mapping scheme will be elaborated in Section IV-C. Initially, *steps* $4, 14$ filter KV caches at a page granularity (i.e., *group* in Algorithm 1), preventing the retrieval of flash pages containing only weak tokens identified by $argtopk$ in *steps* $1, 10$. Subsequently, during *steps* $5, 15$, the NFCs refine the sparse KV caches by passing through only strong tokens from the pages that contain both weak and strong tokens.

**Hardware-Based Attention Engine.** Based on the SparF Attention mechanism, we design the hardware-based SparF engine on the InstCSD and integrate it with the SSD controller. As depicted in Figure 8, the SparF engine is primarily comprised of attention kernels, argtopk unit, and the filters within each NFC. Minor components, such as the summation and normalization units, are omitted in the figure for simplicity.

The workflow of the SparF engine is as follows. To begin with, the $q$ vector of shape $(1 \times d_h)$ is submitted to the `argtopk` unit and filtered to retain only the top-$r$ entries, denoted by $i$. The NFC subsequently uses these top-$r$ indices to retrieve columns $i$ from the $K$ cache in NAND flash with a shape of $(S \times d_h)$, trying to get $K_{[:,i]}$. Despite this selection, the retrieved pages may contain sparse entries due to the size gap between a KV entry (FP16 number, or 2 Bytes) and a flash page (4K Bytes). Therefore, only a subset of sparse columns are filtered out, and we get $K_{[:,i_1]}$. The coarse-grained sparse $K_{[:,i_1]}$ caches are further refined using fine-grained index information (for further details, see Section IV-C). The refined entries, $q_{[i]}$ and $K_{[:,i]}$, are then processed by the `Attention Kernel` to compute an approximate attention score (①). This score is reprocessed through the `argtopk` unit to find out the top-$k$ largest token indices, represented by $j$, which form the final attention output. Based on the indices, the $K_{[j,:]}$ and $V_{[j,:]}$ caches are loaded from flash at the coarse page-level granularity and filtered through the NFC similarly. Specifically, the $q$ and sparse $K_{[j,:]}$ tensors are first loaded and directed to `Attention Kernel` (②). Concurrently, the $V_{[j,:]}$ tensors are loaded in parallel to hide the loading latency. The two instances of `Attention Kernels` in Figure 8 are identical. Each kernel comprises multiple GeMV and Softmax units to complete the attention computation involved in steps $8, 18, 19$ in Algorithm 1. During the execution, both attention kernels can be scheduled for the two attention computations in SparF Attention considering the real-time loads.

### C. Manage and Transmit KV Caches

To facilitate the SparF Attention mechanism within the CSD equipped with flash chips, it is necessary to enable token-indexed random access to the $V$ cache, in other words, in a column-wise manner. Additionally, both token-indexed and channel-indexed random accesses are required for the $K$ cache (i.e., in both column-wise and row-wise manner). Therefore, considering the low storage cost of flash, we opted to store the $K$ matrices twice in different orientations to optimize access efficiency. Additionally, we designed two sets of efficient address mappings with the dual-step loading mechanism. This approach enables random indexing and efficient flash memory access while significantly reducing write amplification.

**Token-Indexed Mapping.** The token-indexed management, or the row-wise manner, is illustrated in Figure 9(a). Specifically, it is worth noting that in mainstream LLMs such as OPT and Llama, each attention head has a hidden size of 128 in FP16 numbers [61], [78]. Since the multi-head attention calculates head-by-head, it implies a minimum reading granularity of 256B. Considering the 4KB page-granularity access of flash, randomly reading the KV cache can lead to performance degradation of up to 16× in conventional FTL.

To address this, we integrate entries of 16 consecutive tokens from $K$ or $V$ cache in the same head into a *group*, identical to the flash page size. Each group consists of 2048 FP16 numbers, which span a subspace of the complete hidden

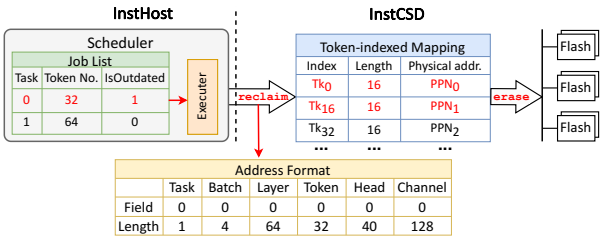



Fig. 10: Illustration of GC mechanism in InstAttention.

embedding corresponding to a specific attention head. For other configurations (e.g., larger attention head or flash page size), the group size varies accordingly. Based on the SparF mechanism, the group may contain a certain number of sparse tokens, which are rows to be ignored. Therefore, during the first loading step, a group will be ignored only if all its tokens rank below the top-$k$ threshold; otherwise, it is considered dense. The sparse tokens remaining in the group are further filtered out through the NFCs, as illustrated in Section IV-B.

To avoid IO conflict and synchronization issues of concurrent accesses, the SparF engine retrieves KV caches in a sequential and pipelined manner. Considering that different token groups within an attention head must be simultaneously loaded for computation, it is essential to maximize throughput by parallelizing retrieval across all flash channels. We therefore stride the KV cache groups across different flash channels in the token dimension. Given that the number of groups accessed per read during the attention computation is substantially greater than the number of available flash channels (typically 4-16), each channel is fully utilized. Our tests imply that this group-based dual-step loading scheme maintains about half sparsity across various datasets during the first-step loading, while the second step reaches full sparsity.

**Channel-Indexed Mapping.** The channel-indexed access, or the column-wise manner, is relatively similar, as illustrated in Figure 9(b). To access entries of consecutive tokens in one hidden embedding channel, we need to store the $K$ cache corresponding to multiple tokens in the same channel within a single page. However, if we still assume the page size as 4KB, each flash page can store 2K entries, which is quite a large granularity for general LLM inference. Therefore, we further adopt the two-step loading mechanism for the channel-indexed access, grouping 2-8 channels into one page. Therefore, the minimum storage granularity in this scenario is 256-1K tokens, which is feasible for both short conversations (less than 256 tokens) and long contexts (longer than 1K tokens). The group size can be dynamically adjusted based on the input length and largest context length of the model in the runtime.

Based on the two mapping schemes, all the computations concerning KV caches are confined within the InstCSD, allowing us to manage KV cache completely in the InstCSD and eliminate the need for a complex host filesystem. InstCSD orchestrates the KV cache data by indexing them with customized logical addresses via the FTL. Specifically, the logical address is defined as a 32-bit integer, which is segmented into multiple fields to uniquely identify the batch, layer, token,

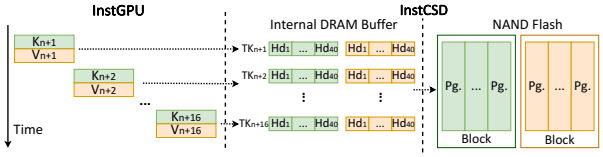



Fig. 11: Batched writing in InstCSD.

head, and channel number of an entry, as illustrated by the *Address Format* in Figure 10. Consequently, the SparF engine directly locates and retrieves the KV cache it needs through two L2P mapping tables, which are stored in the CSD internal DRAM like what traditional SSDs do.

**Batch Writing Requests.** The writing process of KV caches primarily occurs during the prefilling phase, where the entire input sequence is processed in parallel, generating a substantial amount of KV cache. Nevertheless, once all KV cache chunks from the input tokens are transmitted and stored on the flash chips, the decoding phase continues to generate KV vectors for new tokens incrementally. These vectors are written to the CSD in small sizes. As each page contains KV tensors for multiple tokens, the KV caches generated sequentially for these tokens must first be stored in the DRAM group buffer within the CSD, illustrated in Figure 11. These vectors are then flushed back to the flash chips in the background once the DRAM buffer is fulfilled. Furthermore, owing to the mismatch between page-granularity writes and block-granularity erases of NAND flash [24], small write requests can lead to write amplification (WAF) issues, a well-documented challenge for SSDs. To mitigate WAF, it is crucial to ensure that each write operation is at block granularity comprising several hundred pages. Therefore, since the GPU generates new $k, v$ vectors of all attention heads in parallel, enough groups of attention heads can be batched to fill one flash block. For token-indexed KV caches, we prioritize placing groups corresponding to different attention heads within the same block, which can effectively avoid write amplification issues. As all KV caches are sequentially generated and stored in an appending manner, data fragmentation is eliminated within the InstCSD. This obviates the need for foreground GC during writing processes, thereby avoiding any interference (cf. Section IV-D).

### *D. Integrate and Scale the System*

**GPU-CSD coordination.** In InstAttention, the CSD manages the KV cache during the QKV projection process in both prefilling and decoding phases, and calculates attention during decoding. This leads to a pipelined cooperation between the GPU and CSD: In the prefilling phase, the GPU handles all computations are handled. The KV cache for all input tokens is transferred to the CSD via PCIe, a process that may be time-consuming. To mitigate this, we implement a layer-wise pipeline wherein the KV cache generated at the $i$-th layer is transferred to CSD concurrently with the computation at the $(i+1)$-th layer. After attention computation, the CSD sends attention outputs back to the GPU to proceed with generating the $o$ vector and completing the subsequent FFN layer inferences. Compared with the traditional KV cache

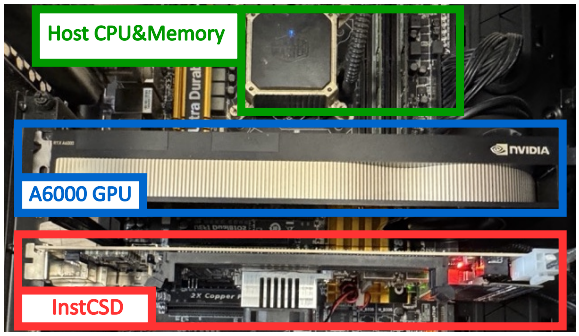


Fig. 12: Hardware deployment of InstAttention.

offloading system, the data volume transmitted on the PCIe buses is reduced by $s/2$, where $s$ refers to the sequence length.

For data transmission between the GPU and the CSD, we use a peer-to-peer approach, bypassing the host memory buffer to enable direct data transfer through PCIe lanes. This approach minimizes redundant data copies and optimizes transmission efficiency. Unlike the traditional GPUDirect Storage [48] approach, which depends on the host filesystem to manage SSD data, InstAttention operates independently of complex host file systems for managing KV cache. Specifically, the host runs InstCSD and InstGPU drivers. In the data plane, the InstCSD driver replaces the logical address field (DWord10 in NVMe commands [14]) with our customized logical address (cf. Section IV-C) to perform `nvme_read()` and `nvme_write()` commands, similar to standard NVMe protocols. The *InstAttention Scheduler* orchestrates data transmission, initiating DMA transactions between the logical addresses of InstCSD and the mapped VRAM address of InstGPU. In the control plane, we leverage the reserved bits in the NVMe command to support three new functions: `config()` to set model hyperparameters, `attend()` to initiate attention computation, and `reclaim()` with specific data address and size to perform GC (Garbage Collection) on InstAttention.

**Garbage Collection.** The garbage collection (GC) process in InstCSD is simplified, which differs significantly from the traditional GC processes observed in SSDs. As the KV cache serves as intermediate activation data, it does not require persistent storage within InstCSD. Instead, GC is periodically initiated by the host scheduler only to erase the stale pages (i.e., obsolete and unnecessary KV cache), thereby preventing the overwhelming of flash capacity. The sequential nature of all KV cache writing requests to InstCSDs, which append rather than modifying existing data, eliminates data fragmentation. This simplification substantially reduces the GC overhead compared to that of traditional SSDs. Furthermore, considering that real LLM serving systems present periodic and fluctuating request intensity [65], GC is invoked only during LLM service intervals when the InstCSD is idle and the available page budget falls below a specified threshold. This approach minimizes interference with attention computation and writing requests. The KV cache data is ordered and erased in an LRU manner, which guarantees the availability of sufficient pages for incoming KV caches.

Figure 10 illustrates an example of the GC process in InstAttention. The host scheduler maintains a job list to record all the inference jobs with their token number, and whether they are outdated. During idle time, the scheduler issues the `reclaim()` command to InstCSD to erase the blocks corresponding to the outdated `Task 0`, and specifies the length to set the range of all the metadata. As we want to erase KV cache data of all the batches and layers of `Task 0` in this example, all the `Fields` are set to zero and `Lengths` are set to the maximum. Upon receiving the command, the InstCSD FTL leverages the two index-based mapping table to find out all the token indices in the specified range of task 0, and executes garbage collection process based on the physical addresses. Note that for simplicity, we only illustrate the token-indexed mapping in the figure, and the channel-indexed mapping follows a similar approach.

**Scale To CSD Array.** InstAttention can be seamlessly scaled across multiple CSDs to significantly improve inference performance. Specifically, the MHA that mainstream LLMs employ allows each head in a multi-head attention layer to be calculated for an independent set of attention scores. As each InstCSD exclusively handles the attention module and different attention heads compute independently without interdependencies, it is feasible to distribute various attention heads across CSDs. For a configuration with $n$ CSDs and $n_{\text{head}}$ attention heads, where typically $n_{\text{head}} \gg n$ (for example, OPT-13B features 40 heads), each CSD processes $n_{\text{head}}/n$ heads. Finally, the outputs from the attention heads processed on different CSDs are transmitted back to the GPU, which then concatenates these results to form the final output.

## V. Implementation

### A. System Deployment

We have implemented InstAttention with real hardware, as illustrated in Figure 12, with full-stack software support. InstCSD is built on Daisyplus OpenSSD, the latest representative NVMe CSD device in the OpenSSD project [31], [60]. It employs a Xilinx ZU17EG MPSoC as its processor, which contains a mid-range FPGA chip with a four-core ARM processor, 2GB DRAM, and PCIe 3.0x4 interface. The SparF engine and NFC filters are implemented on the FPGA part, clocked at a frequency of 285MHz, while the FTL runs as software on the ARM processor. The software stack of InstAttention is built atop FlexGen [57]. Specifically, we consider `TorchDisk` object, which is employed for offloading the KV cache in the original FlexGen implementation, as a `TorchDevice` endowed with computational capabilities. This allows us to leverage the inherent GPU-CPU heterogeneous computing capability, seamlessly integrating the CSD into the established FlexGen framework with the same APIs for users. The driver for InstCSD is customized based on [44], providing simple control and data-plane interfaces (cf. Section IV-D).

| Units | | Latency | Throughput | Accuracy |
|---|---|---|---|---|
| **GeMV** | **Real** | 0.32us | 12.7GFLOPS | 95.50% |
| | **Virtual** | | 13.3GFLOPS | |
| **Softmax** | **Real** | 164us | 14.2MFLOPS | 94.00% |
| | **Virtual** | | 15.1MFLOPS | |
| **Filter** | **Real** | 37us | 1.85GB/s | 96.80% |
| | **Virtual** | | 1.79GB/s | |

TABLE I: Performance and accuracy of InstCSD.

| | LUT(K) | FF(K) | BRAM Tile | DSP |
|---|---|---|---|---|
| Attention Kernel | 99.2 | 207.3 | 96 | 768 |
| Argtopk | 5.83 | 3.87 | 24 | 0 |
| NFC | 58.332 | 27.8 | 96 | 0 |
| NVMe Controller | 7.99 | 12.45 | 27.5 | 0 |
| Interconnect | 4.12 | 6.17 | 7.5 | 0 |
| Available | 218.6 | 437.2 | 545 | 900 |
| **Percent(%)** | 80.27% | 58.92% | 46.06% | 85.33% |

TABLE II: Resource utilization of InstCSD on Zynq7045.

### *B. Towards Practical CSD Solutions*

While OpenSSD serves as a real CSD, it presents several challenges that hinder its widespread adoption. Notably, it features expensive FPGA chips, costing thousands of dollars [60], and is equipped with limited flash resources. Additionally, it only supports legacy motherboards like Z97, which lags far behind the current hardware environment [31]. These specifications fall short of contemporary SSDs with cheaper processors, more channels, and greater storage capacity.

To bridge this gap between experimental setups and practical systems, we adopt NVMeVirt [29], a cutting-edge software-defined virtual NVMe device. NVMeVirt facilitates a seamless integration with the host software stack like a real NVMe SSD with the flexibility to customize SSD internals with specific needs. Therefore, we first collected fine-grained latency statistics of OpenSSD-based InstCSD deployment with the system. We further built the software-defined InstCSD based on the NVMeVirt, setting the corresponding latencies to reflect the speed of InstCSD processing engine on the host CPU. To match real deployment costs, we also prototyped InstCSD on Xilinx Zynq7045 [69], a more economically viable FPGA SoC, prevalently utilized in edge computing. Table I shows the performance statistics of main components in InstCSD for one OPT-13B attention head and 16 tokens, both on the real and virtual CSD devices. We use the evaluated throughput statistics to reflect the emulation accuracies, all of which are around 95% and sufficient to reflect the real system. We further extend the flash channel number to 8, with 1.4GB/s channel bandwidth to align with modern SSD configurations (i.e., Samsung 980pro [55]). The external PCIe interface is 4.0x4, which delivers a maximum of 7GB/s throughput. The detailed resource utilization rates are listed in Table II. We exploit the DSP resources of Zynq7045 to deliver the maximum performance for attention computation.

## VI. Evaluation

### *A. Methodology*

**Inference Systems Setup.** We set seven LLM inference systems to thoroughly evaluate the performance of InstAttention over the current KV cache offloading systems:

1) `DeepSpeed`: DeepSpeed-MII system [21] with Zero-Inference [6]. It represents the latest memory-only KV cache offloading system.
2) `FlexGen`: FlexGen system [57], which represents the latest KV cache offloading system to both host memory and SSD for throughput-oriented scenarios. We configure its offload target to SSD to evaluate the SSD-based offloading scheme.
3) `FlexGen-GDS`: FlexGen with GPUDirect Storage [48];
4) `FlexGen-SparQ`: FlexGen with SparQ Attention for sparsity with 1/8 compression ratio;
5) `Recomp`: vLLM system [32], with recomputation for KV cache when it exceeds the available memory;
6) `InstA`: our baseline InstAttention implementation without the SparF Attention mechanism;
7) `InstA-SparF`: complete InstAttention with SparF Attention for sparsity with 1/8 compression ratio;

**Testbed Configuration.** We conduct our experiments in single CPU-GPU systems. We use NVIDIA A6000 GPU with 48GB VRAM, the 2.2GHz Intel Xeon 5320 CPU with 96GB DDR4 memory, and Samsung 980pro SSDs with 2TB storage. The GPU is connected to CPU via PCIe Gen4x16 lanes.

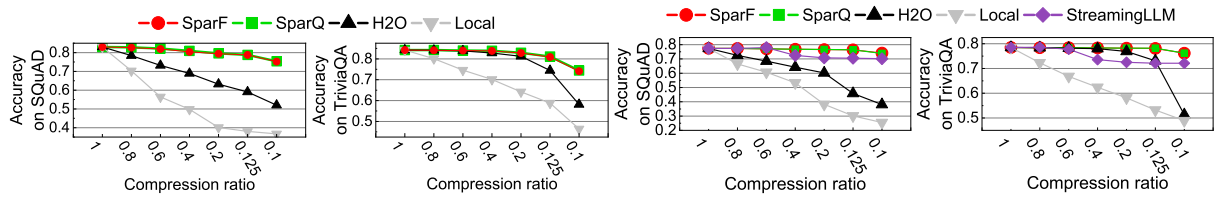
**Model and Datasets.** We evaluate OPT-13B, OPT-30B and Llama-2-13B models, which are representative mid-sized LLMs for resource-constrained scenarios. We extend the original FlexGen to support the latest Llama-2-series models. To accommodate all the parameters of OPT-30B, we use two A6000 GPUs for evaluation. We use FP16 for all variables. The sequences for inference are sampled from popular datasets (i.e., ShareGPT [56], Wiki-Text-2 [45], SQuAD [59], and TriviaQA [27]). For OPT models, both the input and output sequence lengths are set to 1024, while for Llama-2 models they are set to 2048. The configuration matches the maximal context length of OPT and Llama-2 models to fully demonstrate the long-context scenarios with heavy KV cache burden.

### *B. Accuracy*

We evaluate the accuracy of SparF along with the vanilla SparQ Attention, widely-adopted pruning techniques H2O [79] and StreamingLLM [68], and the local attention method, under different KV cache compression ratios, as illustrated in Figure 13a and 13b. As StreamingLLM only supports Llama models, we exclude it from the OPT-model evaluation. We observe that SparF performs nearly identically with the vanilla SparQ Attention while maintaining robustness against H2O, StreamingLLM, and local attention. This is because SparF primarily focuses on optimizing the KV cache access patterns on the flash chips, efficiently identifying the KV cache sparsity with dual-step loading. As SparF suffers negligible accuracy loss with a compression ratio up to 1/8, we set the default compression ratio as 1/8 for the following evaluations.

### *C. Throughput Evaluation*

**Performance with 1 SSD (CSD).** Figure 14 illustrates the end-to-end throughput of different platforms with 1 SSD(CSD) on OPT-13B. `DeepSpeed` leverages host memory for KV




(a) Accuracy on the OPT-13B model.

(b) Accuracy on the Llama-2-7B model.

Fig. 13: Accuracy of different sparsity methods.

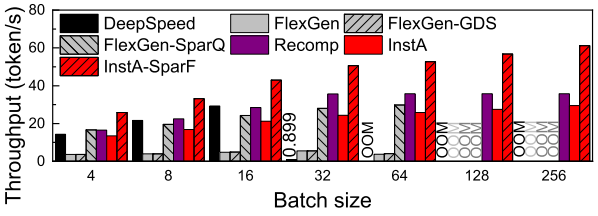



Fig. 14: Throughput of LLM systems: 1-SSD@OPT-13B.

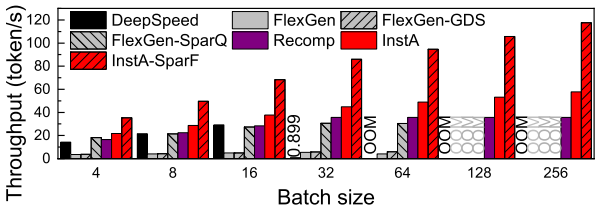



Fig. 15: Throughput of LLM systems: 2-SSD@OPT-13B.

cache offloading, which owns a larger bandwidth and thereby outperforms other dense schemes when batch size is small (4-16). However, it quickly exceeds the available host memory at bs=32 and incurs the kernel swapping to SSDs, which results in a 32.6× throughput degradation compared with bs=16. `FlexGen` supports up to bs=64 but delivers much lower throughput due to the SSD-offloading with larger capacity but limited PCIe bandwidth. Note that the OOM error occurs at bs=128 despite the substantial SSD capacity. This is because the intermediate KV cache during the prefilling phase exceeds the available GPU VRAM. `FlexGen-GDS` gets negligible performance improvement over the original `FlexGen`, because the original GPU-Direct Storage provided by CUDA [48] still relies on host filesystem to manage data on the SSD.

In contrast, `InstA` bypasses the host filesystem, enabling direct access to the KV cache on InstCSD. `InstA` also leverages a layerwise transmission of the KV cache during the prefilling phase, which significantly reduces the VRAM buffer requirement for the intermediate KV caches. Therefore, `InstA` supports much larger batch sizes, and addresses the bandwidth challenges in traditional offloading systems, thereby outperforming `FlexGen` by 6.85× at bs=64. `InstA` shows the best scalability as batch size increases. Note that `InstA` only outperforms the maximal achievable throughput of `DeepSpeed` (at bs=16) by 4.6%, because the CSD internal bandwidth (11.2GB/s) is still lower than the PCIe bandwidth between GPU and host memory (32GB/s). `InstA-SparF` effectively reduces the demanding KV cache volume, which further improves the throughput of original `InstA` by up to 2.08× at bs=256, outperforming the baseline `FlexGen` by up to 11.1×. Lastly, `Recomp` achieves the best performance at small batch sizes. Nevertheless, for large batch sizes, `InstA-SparF` still outperforms `Recomp` by up to 71.3% (at bs=256). `Recomp` is limited by enormous KV cache recomputation on large batches, and incompatible with sparse KV cache techniques such as SparF.

**Performance with 2 SSDs (CSDs).** Figure 15 further presents the throughput with 2 SSDs(CSDs). We observe that traditional offloading schemes exhibit negligible performance improvement despite larger PCIe bandwidth aggregated by multiple SSDs. This is because these schemes rely on the host filesystem to manage KV cache on the SSD, which puts a heavy burden on the data transmission between GPU and SSD. `InstA` addresses this issue through two approaches. On the one hand, the optimized P2PDMA transmission between GPU and CSDs bypasses the host; on the other hand, most of the KV cache transmission occurs within the CSD through the internal flash channels, which can be easily scaled up through multiple CSDs. Therefore, `InstA` (at bs=256) outperform maximal achievable throughput of `FlexGen` (at bs=32) by 10.5×, and `InstA-SparF` (at bs=256) outperforms `FlexGen-SparQ` (at bs=32) by 3.11×, respectively.

**Performance on other models.** To illustrate the potential and scalability of InstAttention, we further evaluate the OPT-30B and Llama-2-13B models, as shown in Figures 16 and 17, respectively. For OPT-30B, `InstA` and `InstA-SparF` (at bs=128) outperform `FlexGen` (at bs=8) by up to 4.09× and 9.39×, respectively, exhibiting performance advantages similar to the 13B model. For the Llama-2-13B model with 4K context, `InstA` and `InstA-SparF` (at bs=128) outperform `FlexGen` (at bs=16) by up to 5.68× and 12.48×. Note that `Recomp` also shows obvious performance advantages, outperforming `FlexGen` by up to 9.26×. Nevertheless, since InstAttention adopts the bandwidth-efficient sparsity mechanism and can be further enhanced by aggregating multiple CSDs (cf. Figures 15 and 20a), thereby showing greater potential in offline long-context inference.

### *D. Latency Breakdown*

**Decoding Latency Analysis.** Figure 18 and 19 depicts the normalized latency breakdown during the decoding phase of different LLM systems, respectively, where `InstA-2` InstAttention with 2 CSDs. Both sparse (1/8) and dense attention are evaluated, with small (bs=4), middle (bs=64), and large batch size (bs=256) scenarios. We observe that the KV cache

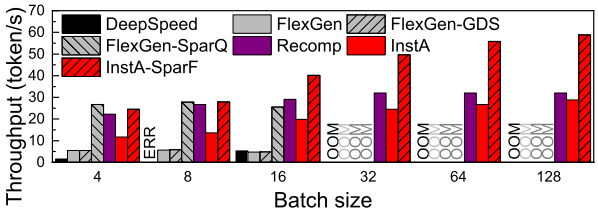



Fig. 16: Throughput of LLM systems: OPT-30B.

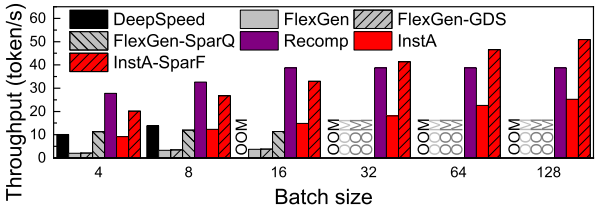



Fig. 17: Throughput of LLM systems: Llama-2-13B.

access is the primary bottleneck across all the scenarios and systems, considering the extremely low arithmetic intensity of the attention computation. Nevertheless, compared with `FlexGen` at bs=64, `InstA` and `InstA-2` still reduce the KV cache access percentage from 98.9% to 80.7% and 76.4% in dense inference, and from 92.4% to 82.3% and 74.0% with sparsity, respectively. To further alleviate the bottleneck, it is promising to scale up to more CSDs or flash channels.

**SparF Attention Engine Analysis.** We dive into the SparF Attention engine in InstCSD to analyze the normalized overheads of each unit, as illustrated in Figure 21. Compared with dense attention computation, the primary difference lies in that SparF introduces an additional `Logit-0` process, corresponding to the step 4 in Algorithm 1. The extra logit computation further helps identify the sparsity within the sequences, which finally delivers the overall performance improvement.

### *E. Scalability And Sensitivity Tests*

**Scalability with More CSDs.** We further evaluate the scalability of InstAttention with more CSDs in terms of both dense

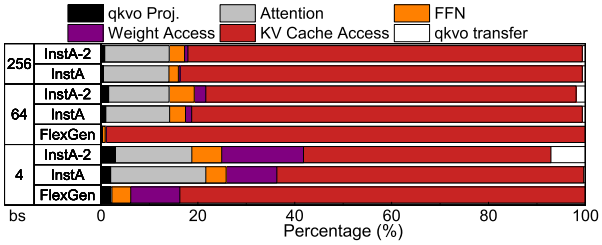



Fig. 18: Latency breakdown of dense LLM inference.

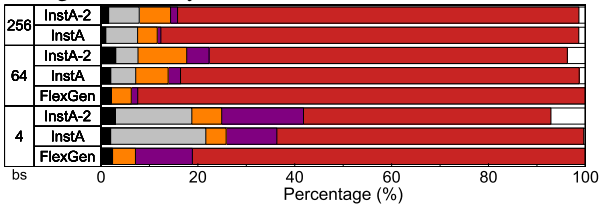
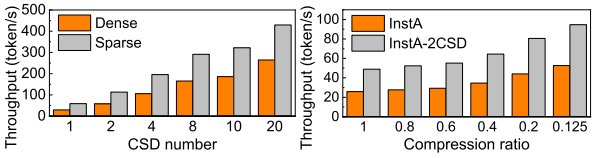



Fig. 19: Latency breakdown of sparse LLM inference.




(a) Varying number of InstCSDs. (b) Varying compression ratios.

Fig. 20: Throughput with varying configurations.

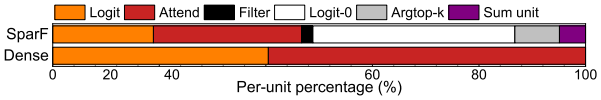



Fig. 21: Latency breakdown of the SparF Attention engine.

inference and 1/8-sparsity at bs=256, as depicted in Figure 20a respectively. Since traditional KV cache-offloading systems with SSDs show negligible performance improvements scaling with SSD number, we omit them in the Figure. Compared with 1-CSD configuration, 20 CSDs can improve the dense and sparse inference throughput by 8.99× and 7.29×, respectively. The head-level parallelism is employed among multiple CSDs, which is suitable for InstAttention because only the critical attention computation and KV cache are offloaded. As these computations and data are inherently parallel and have no dependency, the scaling up can be quite straightforward by assigning attention heads to multiple CSDs. Therefore, both the dense and sparse (with SparF) `InstA` show good scalability with an increasing number of CSDs.

**Sensitivity with Varying Sparsity.** Figure 20b shows the throughput of `InstA` with 1 or 2 CSDs under different compression ratios with SparF Attention, respectively. We observe that although a larger compression ratio leads to more random fine-grained access to KV cache on the flash chips, which is typically a challenge for SSDs, InstAttention efficiently benefits from larger compression ratios due to the efficient dual-step loading mechanism of SparF Attention.

### *F. Overhead Analysis*

**Endurance Analysis.** Although InstAttention involves a substantial amount of KV cache writing to flash, potentially causing severe endurance issues, we contend that modern SSDs are capable of prolonged inference tasks. We use the V-NAND V6 NAND flash as a reference model, which is the NAND chip utilized in the Samsung 980 Pro SSD, featuring 3,000 P/E cycles [55]. Note that the assumed theoretical endurance is larger than the original 980 Pro model, which is attributed to the simplified FTL of InstAttention to avoid extra GC or wear-leveling processes. This minimizes write amplifications and thereby contributes to the optimal endurance of flash chips. Therefore, for an InstAttention instance equipped with four dedicated CSDs and targeting a 13B model, which generates approximately 0.78MB of KV cache per token, the system can accommodate 32,263,877K tokens. Furthermore, given that KV caches represent intermediate activation data that are either discarded or refreshed shortly, reducing the NAND flash retention time from the typical three years to three weeks—which is adequate for LLM inference tasks—would

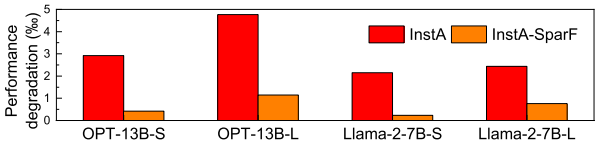
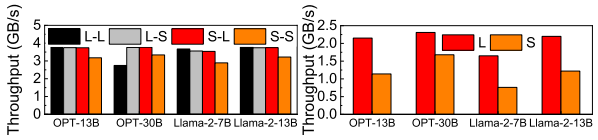



Fig. 22: Degradation with the worst read-retry (‰).




(a) Prefilling phase. (b) Decoding phase.

Fig. 23: Writing performance of InstCSD in different phases.

increase the endurance (i.e., P/E cycles) by a factor of approximately 6.67×, as indicated by prior studies [1], [7], [41]. Consequently, assuming a user with extremely high-intensity use of LLM inference service consumes 128K tokens per day, the actual serving capacity could accommodate at least 920 extreme users over five years, 1530 users over three years, or 4600 users over one year. Considering that typical commercial SSDs provide a warranty period ranging from 3 years to 5 years [7], this serving capacity is sufficient for a resource-constrained LLM server.

Considering the possible read-retry exacerbation due to the reduced $V_{TH}$ margin [51] for modern TLC-based SSDs, the recent research has revealed promising advancements in this area. To be specific, by modeling the characteristics of flash chips and predicting the optimal read voltage offset, the state-of-the-art approaches [73] can establish a dynamic read retry table for NAND flash, which has been demonstrated to achieve near-zero read retries. Therefore, based on this study, we assessed the performance of InstAttention under various workloads by taking the read retry statistics (up to 5 retries for a single page read request, average 0.003 retries) in the worst case (i.e., after 8K P/E cycles and 10 days of baking at 85°C) from the prior work [73] as the estimated value for InstCSD after retention relaxation. We further tested the performance of InstAttention across various workloads after the worst-case-based read retry scenario, as illustrated in Figure 22. The `-S` or `-L` tag represents inference with Small batch size (4) or Large batch size (64). The experimental results indicated that the performance degradation of InstAttention was limited to a maximum of 4.7‰, and `InstA-SparF` shows more negligible penalty due to less KV cache reading from flash chips. These findings suggest that, for InstCSD, leveraging retention relaxation to trade retention time for enhanced endurance is a feasible approach.

**SSD IO Analysis.** Although NAND flash exhibits significantly worse writing performance compared to reading one, we contend that in the InstAttention architecture, writing requests have a minimal impact on performance penalties during long-context LLM serving. Table III details the accumulated I/O volume transferred between InstGPU and InstCSDs, which are collected from evaluations on the OPT-13B model at a batch size of 64. The data read during the decoding phase is

| | IO volumes (GB) | Throughput (GB/s) |
|---|---|---|
| **Prefilling read** | 0 | 0 |
| **Prefilling write** | 1.34 | 3.78 |
| **Decoding read** | 1085.65 | 7.66 |
| **Decoding write** | 1.93 | 2.15 |

TABLE III: IO on OPT-13B to InstCSD during inference.

approximately 810× and 560× greater than the data written during the prefilling and decoding phases, respectively. This considerable reading traffic stems from the repetitive decoding process, where each token generation necessitates reading all existing KV caches from the NAND flash.

We further conduct more detailed analysis on the writing performance of InstCSD with diverse workloads, as illustrated in Figure 23a and Figure 23b for prefilling and decoding phase, respectively. For prefilling phase, we tested both large (64) or small (4) batch sizes (represented by the first `L/S` tag in the figure), and long (2K) or short (128) sequences (represented by the second `L/S` tag). For decoding phase, as the writing granularity is always one token, we only tested different batch sizes. In most cases except `OPT-30B L-L`, the writing data volume is relatively minor and can be buffered by the InstCSD internal DRAM. Therefore, the poor writing speed of NAND flash does not compromise the overall system performance. For decoding phase, the writing performance shows a slight degradation due to the small writing granularity (1 token). Considering the small writing traffic volume, the overall performance penalty is still negligible. These observations indicate that the primary I/O overhead is attributed to reading during the decoding phase, whereas the impact of writing on LLM inference performance is minimum.

## VII. Related Works And Discussion

**PIM-Based Transformer Acceleration.** Several works [11], [20], [67], [81] explore Processing-In-Memory techniques to address the storage and bandwidth bottleneck of LLM inference, which integrate computing units within the memory cells for the memory-bound attention computation. However, these works are all based on simulators, considering that PIM devices are still expensive and far from being widely deployed in practice, especially for resource-constrained scenarios. In contrast, InstAttention is deployed in real hardware, adopting economical CSDs as a more cost-effective and scalable solution to address the storage and bandwidth challenges.

**High-bandwidth Chip Interconnect Solution.** Prior works such as the Nvidia GH (Grace-Hopper) chip [13] feature high-bandwidth inter-chip interconnect between GPU and CPU to address the LLM bandwidth limitation. However, compared with InstAttention, GH chips are cost-prohibitive at about $30,000 in contrast to the more affordable CSDs like Samsung's SmartSSD at approximately $1500 [58], limiting their use in resource-constrained scenarios. Moreover, GH chips support a maximum of 624GB of fast-access memory, insufficient for extensive KV cache demands for scenarios such as multi-turn conversations or memorization. Therefore, we believe both GH chip and InstAttention contribute to boosting long-context LLM inference from different aspects

(i.e., throughput and capacity). Combining GH chips with InstAttention could be a promising solution in the future.

**Optimizations For KV Cache Management.** vLLM [32] manages KV cache in GPU VRAM and host memory in block-granularity, which takes inspiration from the virtual memory mechanism, to reduce the overhead of fragmentation. LMDeploy [25] and CachedAttention [16] focus on managing KV caches on the host memory and SSDs to reduce the recomputation overheads in multi-turn conversations. These works aim to optimize the prefilling phase in online inference scenarios. However, they are not suitable for inference with long output sequences. Other solutions [19], [33], [52], [53], [80] leverage disaggregated resources (i.e., GPU, CPU and memory pools) to store the KV cache and accelerate long-context LLM inference, which are not suitable for resource-constrained scenarios. InstAttention leverages cost-effective CSDs, which are more applicable and effectively address the decoding-phase bottleneck of KV cache.

**Insights For Non-CSD System.** Although InstAttention is primarily designed to offload KV cache and attention to CSDs, traditional non-CSD systems can still benefit from InstAttention. With a specifically calibrated KV cache management system on the host CPU to reduce random and small accesses to the SSDs, we believe a host-side SparF engine can also boost the inference performance with SSD-based offloading.

## VIII. Conclusion

In this work, we introduced InstAttention, a novel CSD-based LLM offline inference system to address the substantial storage and bandwidth challenges associated with KV caches in a cost-effective approach. By offloading the critical decoding-phase attention and KV cache to CSDs with flash-aware designs, InstAttention exploits high channel bandwidth of flash chips, circumventing the limitations imposed by external PCIe bandwidth. Our evaluation shows that InstAttention outperforms current SSD-offloading systems by up to 11.1× for long-context inference in resource-constrained scenarios.

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